Product Requirements Document (PRD)

Project: Internal Customer Success Dashboard

Role: Product Analyst Assignment

Candidate: Prathmesh Aphale

1. Problem Statement

Currently, the ClearQuote Customer Success (CS) team relies on a fragmented system of multiple spreadsheets and manual reports to track customer health, usage trends, and support activity. This decentralized approach creates data silos, increases the time required for CSMs to prepare for client syncs, and makes it difficult for leadership to identify churn risks or upsell opportunities at a glance.

Objective: Design and build a centralized, "single-pane-of-glass" internal web dashboard that aggregates Customer, Usage, Support, and Fleet data into an actionable, easy-to-digest format.

2. User Personas

- **Customer Success Manager (CSM):** The primary user. Needs to quickly investigate specific accounts before meetings, track month-over-month product adoption (API usage, inspections), and ensure critical support tickets for their assigned accounts are not stalled.
- **Head of Customer Success:** The secondary user. Needs an aggregate view of total MRR, global fleet composition, and overall ticket resolution health to allocate team resources and identify macro-trends across the user base.

3. Requirements & Prioritization

For this Minimum Viable Product (MVP), requirements were prioritized to ensure core workflows were supported within the 3-4 hour development window.

P0: Must Have (Core MVP Functionality)

- **Data Integration:** Ability to ingest static data from the 4 core domains (Customers, Usage, Tickets, Fleet).
- **Multi-Tab Layout:** Logical separation of data into four distinct views to prevent cognitive overload.
- **Actionable Support View:** Direct visibility into open, in-progress, and escalated tickets, filtered from resolved historical tickets.
- **Usage Filtering:** Ability to filter usage metrics down to a specific Customer ID / Company Name.

P1: Should Have (Enhanced Usability)

- **High-Level KPIs:** "At-a-glance" metric cards for total customers, total MRR, and churn count.
- **Geospatial Data:** A map view mapping customer locations to understand regional concentration.

P2: Nice to Have (Future Iterations)

- **Google SSO:** Restricting access to @clearquote.io emails. *(Deprioritized for this local MVP, but the Streamlit architecture supports easy integration with OAuth/SSO middleware in a deployed environment).*
- **Live CRM Sync:** Replacing static .xlsx ingestion with live API connections to Salesforce/HubSpot and Zendesk.

4. Data Modeling & Architecture

Tech Stack Selection: Python & Streamlit Streamlit was chosen over a traditional React/Node.js stack to optimize for development speed and native data-handling capabilities. Streamlit seamlessly integrates with Pandas and Plotly, allowing for rapid transformation of Excel data into interactive visual components.

Data Handling & Joins:

- Data is currently ingested via the openpyxl engine directly from the provided multi-sheet workbook.
- **Usage Aggregation:** When viewing global data, usage metrics are aggregated (grouped by Month) using sum and mean functions to provide an accurate macro-view.
- **Calculated Metrics:** Dynamic recalculation is built in (e.g., recalculating global Damage Rate (%) dynamically when viewing the aggregate dashboard rather than relying on static column averages).

5. Design Decisions & Trade-Offs

1. **Focus on "Actionability" over "Data Dumping":** On the *Support & Comms* tab, I explicitly filtered the raw data table to exclude "Resolved" tickets. CSMs do not need a list of past successes when triaging; they need to see exactly what requires their immediate attention.
2. **Aggregated vs. Granular Usage:** I built the *Usage Metrics* tab to default to a macro-level view of all customers, but included a select-box filter. This allows the dashboard to serve both the Head of CS (looking at global trends) and the individual CSM (drilling down into a specific account) without needing two separate dashboards.

3. **Static MVP Limitations:** Because this app runs on static Excel data, there is no write-back capability. In a production environment, this dashboard would serve purely as a read-only presentation layer, with actual data mutation occurring in the underlying CRM or ticketing systems.